

Results: At last follow-up, 17 patients in the re-irradiated group survived without metastasis, 1 was alive with metastasis, 10 were dead of metastasis and 3 of other causes. In the enucleated group, 15 patients were alive without disease, one was alive with metastasis, 25 were dead of metastasis and 2 of other causes. Median survival duration in enucleated and re-irradiated groups was, respectively, 42 and 90 months ($p = 0.040$, log-rank test). Median time free of metastases was 38 months in enucleated and 97 months in re-irradiated patients ($p = 0.028$, log rank-test). At 5 years after enucleation or re-irradiation, the probability of overall survival was, respectively, 36% and 63%; the probability of freedom from metastases was, respectively, 31% and 66%. However, the differences in risk of all cause mortality and of onset of metastasis were not significant after adjustment for recurrent tumor volume in the proportional hazards model. Nine eyes had been enucleated in the irradiated group, 5 for local recurrence and 4 because of intractable pain. The probability of survival with eye retention at 5 years was 40% for the re-irradiated group.

Conclusions: This retrospective analysis suggests that survival in re-irradiated patients is not compromised by administration of a second course of PBRT for recurrent uveal melanoma. The study also shows that patients with uveal melanoma recurrent after high dose fractionated proton beam therapy can be re-irradiated with the same modality using the same dose as given initially, with acceptable morbidity, and with a reasonable expectation of eye retention.

1092 Effectiveness of Fractionated Stereotactic Radiation Therapy for Uveal Melanoma

K. Muller,¹ G. Luyten,² P. Nowak,¹ J. Marijnissen,¹ C. de Pan,¹ P. Levendag¹

¹Radiotherapy, Erasmus MC, University Medical Center, Rotterdam, Netherlands, ²Ophthalmology, Erasmus MC, University Medical Center, Rotterdam, Netherlands

Purpose/Objective: Uveal melanoma has traditionally been treated by enucleation. Since 1970, various eye sparing treatment modalities have been developed with as aim an optimal sparing of visual function. Brachytherapy appears to be a good alternative for enucleation in the treatment of small uveal melanoma, according to recent comparative studies. For medium sized tumors, proton beam and helium ion radiation techniques are widely used. Radiosurgery for uveal melanomas has been introduced in 1987 with excellent results of local control. The most important disadvantage of single fraction high dose radiosurgery is an enhanced complication probability, which causes the loss of radiobiological sparing if critical structures are adjacent to the tumor. Fractionated stereotactic radiotherapy reduces the effective dose in the surrounding tissues with a delivery of an equivalent effective dose to the tumor.

This study analyses the effectiveness and side effects of fractionated Stereotactic Radiation Therapy (f.SRT) for uveal melanoma in a prospective, nonrandomized, self-controlled clinical trial.

Materials/Methods: Between 1999 and 2003, 38 uveal melanoma patients (21 male, 17 female) with mean tumor thickness of 6.4 mm were included in the study. Informed consent was obtained from all patients. The mean follow-up was twenty-five months, with a minimal follow-up of 10 months. No patients were lost to follow-up. A total dose of 50 Gy, prescribed to the 80% isodose line, was given in 5 daily fractions of 10 Gy by LINAC-based SRT. A blinking light in front of the unaffected eye and a camera to monitor the position of the diseased eye are fixed to a non-invasive relocatable stereotactic frame. The main primary outcome measures were local tumor control and visual acuity at 3, 6, 12 and 24 months respectively and acute toxicity (peri- and intraocular side-effects).

Results: After three months (38 patients) the local control was 100%, after 12 months (32 patients) one patient was treated by transpupillary thermotherapy because the tumor size did not decrease and after 24 months (15 patients) no recurrences were seen. The visual acuity gradually declines from a mean of 0.21 at diagnosis to a mean of 0.06 two years after therapy. The acute side effects after three months were: conjunctival symptoms (10), loss of eye lashes (6), visual symptoms (5), fatigue (5), dry eye (1), cataract (1), and pain (4). One eye was enucleated 2 months post f.SRT.

Conclusions: Preliminary results demonstrate that fractionated SRT is an effective and safe treatment modality for uveal melanoma with excellent local control and mild acute side effects. The follow-up period is too short to discuss meaningfully the risk for long term side effects.

1093 Decreased Incidence of Neovascular Glaucoma by Sparing Anterior Structures of the Eye for Proton Beam Therapy of Ocular Melanoma

R. R. Rajendran,¹ V. Weinberg,¹ I. K. Daftari,¹ J. R. Castro,¹ J. M. Quivey,¹ D. H. Char,^{2,3} T. L. Phillips¹

¹Department of Radiation Oncology, University of California San Francisco, San Francisco, CA, ²The Tumori Foundation, San Francisco, CA, ³Department of Ophthalmology, Stanford University, Stanford, CA

Purpose/Objective: We previously hypothesized that sparing the anterior segment of the eye would reduce the incidence of neovascular glaucoma (NVG) (Daftari, I. K., et al., IJROBP 1997, 39:997-1010; Meecham, W.J., et al., Arch Ophthalmol 1994, 112:197-203). To test this hypothesis, we examined the incidence and contributing factors of NVG in patients treated with proton beam therapy at the Crocker Nuclear Laboratory faculty in Davis, California using techniques for sparing anterior structures of the eye.

Materials/Methods: Using anterior structure sparing techniques, 346 patients were treated with proton beam therapy between June 1996 and November 2003 and included in the ocular melanoma database of the Tumori Foundation. Twenty-four patients were excluded due to prior surgical resection and 3 patients because they were treated for recurrences after ¹²⁵I plaque therapy. This resulted in 319 evaluated patients treated with anterior structure sparing techniques. The patients in the study received 56 GyE in 4 fractions. The primary endpoint of this study is the occurrence of NVG. The median follow-up is 24.7 months. The log rank test was used to compare distributions of the occurrence of NVG. The likelihood ratio test (LLRT) from Cox's proportional hazard model was used to compare models to identify independent predictors of outcome.

Results: NVG occurred among 39 of the patients. The probability of remaining free of NVG at 3 years was 89%. The smaller the tumor diameter was, the better the outcome with 3 year estimates of NVG for the quartiles <7.5 mm, >7.5 mm - 10.5 mm,